Markov Random Fields with Spatial Seeds

Bayesian Estimation on Hierarchical Hidden Field Model

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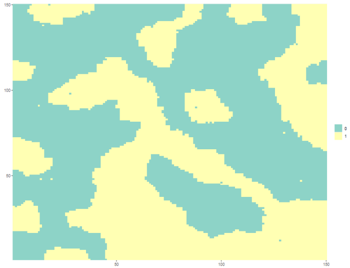
2026-09-17

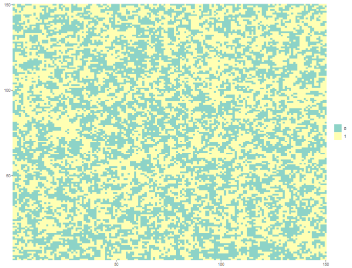
1. Introduction

The Potts Model

- Spatial models on regular lattices (pixels) often employ **Markov Random Fields (MRF)**.
- The **Potts Model** is the classical standard for capturing local neighborhood dependence.



(a) High neighborhood dependence



(b) Low neighborhood dependence

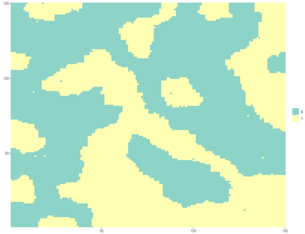
The Potts Model

- **Energy Function:** For the traditional Potts Model, we define a spatial field Z where the local probability of pixel i taking class k is proportional to $exp(\eta_i(k))$:

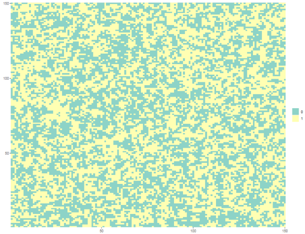
$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0)$$

- **Parameters Meaning:**
  - $N(i)$**:** Pixel i neighbors.
  - α **(Uniformity):** Local attraction between neighboring pixels (smoothing).
  - β **(Global Abundance):** Force favoring/penalizing the background (class 0).

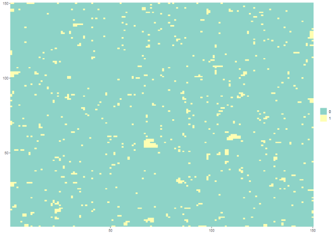
The Potts Model



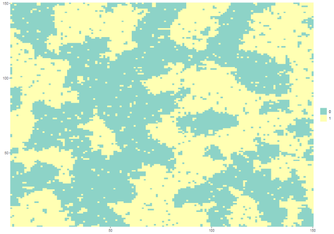
(a) $\alpha = 2, \beta = 0$



(b) $\alpha = 0.5, \beta = 0$



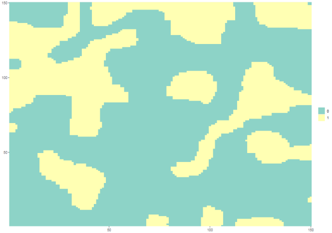
(c) $\alpha = 1, \beta = 0.1$



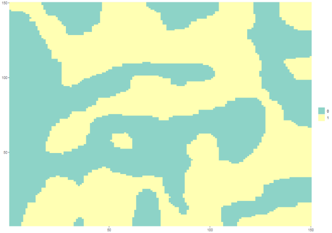
(d) $\alpha = 1, \beta = 0$

The Spatial Limitation

- Many real images and textures possess spatial patterns we call **"seeds"** that may attract or repel colors.
- **Our Proposal:** Incorporate spatial seeds that exert radial influence on the classes.
- **Objective:** Propose a reparameterized model that describe the inclusion of centroids/seeds, and validate its inference under a hidden model.



(a) $\alpha = 3, \beta = 0$



(b) $\alpha = 3, \beta = 0$

2. The Proposal

Potts Reparameterized with Seeds

- **Enhanced Local Energy Function:** To address the isotropic limitation of the Potts model, we introduce a few elements to handle the spatial centrality bias in our local energy funtion:

$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0) + \sum_s \frac{\delta_s}{1 + d(i, s)} \mathbb{1}(c_s = k)$$

- **Parameters Meaning:**
  - α **(Uniformity):** Local attraction between neighboring pixels (smoothing).
  - β **(Global Abundance):** Force favoring/penalizing the background (class 0).
  - δ_s **(Centrality):** Radial attraction/repulsion of the seed s.
  - $d(i, s)$**:** Distance between pixel i and seed s.
  - c_s**:** Class/color related to the seed.

Potts Reparameterized with Seeds

- **Enhanced Local Energy Function:**

$$\eta_i(k) = \alpha \sum_{j \in N(i)} \mathbb{1}(Z_j = k) + \beta \mathbb{1}(k = 0) + \sum_{s} \frac{\delta_s}{1 + d(i,s)} \mathbb{1}(c_s = k)$$

- **Enhanced Global Energy Function:**

$$U(\omega) = \alpha \sum_{\{i,j\} \in E} \sum_{k \in S} \mathbb{1}(\omega_i = k) \mathbb{1}(\omega_j = k) + \beta \sum_{i \in N} \mathbb{1}(\omega_i = 0) + \sum_{i \in N} \sum_{s} \frac{\delta_s}{1 + d(i,s)} \mathbb{1}(\omega_i = c_s)$$

The Threshold of the "Area of Influence"

- **Equilibrium of Forces:** The local seed attraction competes against the global abundance of the background class.
- **Null Boundary:** Under our model, ignoring neighborhood smoothing, they cancel out when:

$$\eta_i(0) = \eta_i(k) \implies d_{i,s} = \frac{\delta_s}{\beta} - 1$$

- **Spatial Transition:** That "Null Boundary" radius gives to the model a rough outline of the objects in the image.
  - **Within radius d_i:** The seed attraction dominates the local probability.
  - **Beyond radius d_i:** The class 0 dominates.
  - **Smoothing effect:** Depending on α value, pixels can be affected more by its neighbors than by the radius rule.

The Theoretical Bottleneck: Intractable Likelihood

- **The Gibbs Measure:** The global probability of an image configuration ω is governed by:

$$P(\omega) = \frac{1}{Z(\theta)} \exp\{U(\omega)\}$$

- **The Computational Bottleneck:** The term $Z(\theta)$ is the **normalizing constant** (partition function) summed over all lattice configurations Ω:

$$Z(\theta) = \sum_{\omega \in \Omega} \exp\{U(\omega)\}$$

- For a small 50×50 grid and $K = 3$ colors, this requires 3^{2500} terms, rendering standard Maximum Likelihood impossible.

3. Hidden & Hierarchical Models

The Major Evolution: The Hidden Model (HMRF)

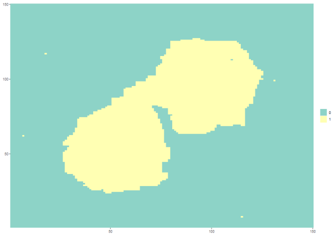
- **The Latent Field:** In segmentation practice, the label field Z (the true texture) is **latent/hidden**.
- **Observed Intensities:** We only observe the noisy, continuous intensities Y.
- **Observation Equation (Emission):** Conditionally on the latent label $Z_i = k$, the intensity Y_i is Normal:

$$Y_i \mid Z_i = k \sim N(\mu_k, \sigma^2)$$

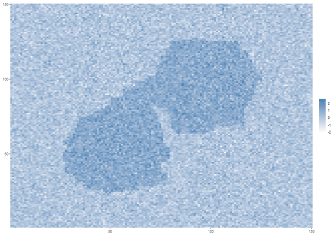
- **Normal Mixture:** Marginalizing over Z_i, Y_i forms a spatial mixture of Gaussians:

$$f(y_i) = \sum_{k=0}^{K-1} P(Z_i = k)\phi(y_i; \mu_k, \sigma^2)$$

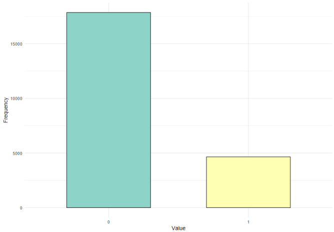
The Major Evolution: The Hidden Model (HMRF)



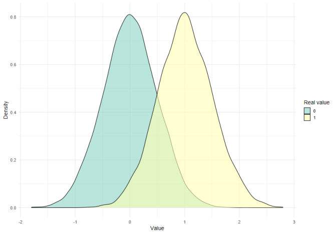
(a) 2-seeds generated image



(b) Same image plus $N(0, 0.5)$ noise



(c) Clean image color frequencies



(d) Noisy image color densities

Hierarchical Bayesian Formulation

- **Full Joint Distribution:** Treating Z, parameters θ, and seed coordinates S as random variables:

$$P(Y, Z, \theta) = P(Y \mid Z, \mu, \sigma^2) P(Z \mid \alpha, \beta, \delta, S) \times \\ \pi(\alpha)\pi(\beta) \prod_{j=1}^{M} \pi(s_j)\pi(\delta_j) \prod_{k=0}^{K-1} \pi(\mu_k)\pi(\sigma^2)$$

- **Prior Distributions:**
  - **Field (Gamma):** $\alpha \sim \text{Gamma}(a_\alpha, b_\alpha)$, $\beta \sim \text{Gamma}(a_\beta, b_\beta)$.
  - **Means (Normal):** $\mu_k \sim N(m_k, \tau_k^2)$ (anchored via K-means).
  - **Variances (Inverse-Gamma):** $\sigma^2 \sim IG(a_\sigma, b_\sigma)$.
  - **Spatial:** Seed position $s_j \sim \text{Unif}(L)$ discretized on the grid.

4. Estimation & Algorithms

The Metropolis-Hastings Step via Pseudoposterior

- **Fixing the Bottleneck:** Because the true likelihood is intractable, we replace $P(Z \mid \theta)$ with the Pseudolikelihood $PL(\theta; Z)$ in the Metropolis-Hastings acceptance step.

$$P(Z_i = k \mid Z_{-i}, \theta) = \frac{P(Z_i = k, Z_{-i} \mid \theta)}{P(Z_{-i} \mid \theta)} =$$

$$\frac{\frac{1}{Z(\theta)} \exp\{U(Z_i = k, Z_{-i})\}}{\sum_{\ell=0}^{K-1} \frac{1}{Z(\theta)} \exp\{U(Z_i = \ell, Z_{-i})\}}$$

$$PL(\theta; Z) = \prod_{i \in N} \frac{\exp\{\eta_i(z_i)\}}{\sum_{\ell=0}^{K-1} \exp\{\eta_i(\ell)\}}$$

- **The Target Pseudoposterior:** This substitution formally defines our target distribution as the **Pseudoposterior**:

$$\pi_{\text{pseudo}}(\theta \mid Z) \propto PL(\theta; Z)\pi(\theta)$$

The Metropolis-Hastings Step via Pseudoposterior

- **Acceptance Probability:** For a symmetric proposal distribution $D(\theta^* \mid \theta)$, a proposed parameter vector θ^* is accepted with probability $\min(1, A)$, where:

$$A = \frac{\pi_{\text{pseudo}}(\theta^* \mid Z)}{\pi_{\text{pseudo}}(\theta \mid Z)} = \frac{PL(\theta^*; Z)\pi(\theta^*)}{PL(\theta; Z)\pi(\theta)}$$

- In log-scale, the proposal is accepted if:

$$\log(U) < \log PL(\theta^*; Z) + \log \pi(\theta^*) - \log PL(\theta; Z) - \log \pi(\theta)$$

The Metropolis-within-Gibbs Algorithm

0. **Initialization**: Noise parameters (μ_k, σ^2) are calculated via K-means over the observed intensity Y_i and they are used among the spatial parameters prioris to generate the initial $Z^{(0)}$ hidden field.
1. **Step A (Metropolis-Hastings - Field):** Propose (α, β, δ) via a random walk, evaluated over the **Pseudolikelihood (PL)** (but can also be initialized by user-chosen prioris).
2. **Step B (Metropolis-Hastings - Seeds):** Propose discrete 1-pixel steps for the coordinates (x, y) of the seeds.
3. **Step C (Gibbs - Latent Field):** Sample $Z_i^{(t)}$ conditioned on its neighbors and the observed intensity Y_i.
4. **Step D (Gibbs - Noise):** Sample new (μ_k, σ^2) directly from closed-form analytical distributions.

Full Conditionals and Conjugacy

- **Updated Posterior Mean (μ_k):** The full conditional under conjugate priors is a Normal:

$$m_{\text{post}} = \frac{\frac{m_k}{\tau_k^2} + \frac{\sum_{i:Z_i=k} y_i}{\sigma^2}}{\frac{1}{\tau_k^2} + \frac{n_k}{\sigma^2}}$$

$$\tau_{\text{post}} = (\frac{1}{\tau_k^2} + \frac{n_k}{\sigma^2})^{-1}$$

- **Updated Posterior Variance (σ^2):** Follows an Inverse-Gamma:

$$a_{\text{post}} = a_\sigma + \frac{n_k}{2}$$

$$b_{\text{post}} = b_\sigma + \frac{1}{2} \sum_{i:Z_i=k} (y_i - \mu_k)^2$$

Code Engineering & Stability Challenges

- **Rcpp for Real-World Grids:** Recreated the local neighborhood and distance calculations in **C++ via Rcpp** to make iterations highly efficient.

5. Application

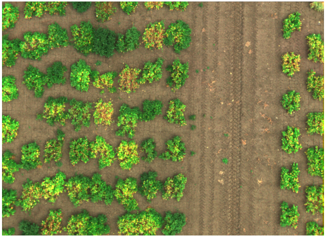
Image Segmentation Application

- **The test:** In order to evaluate our model in real world data, a near-infrared image by Aslahishahri et al. (2021) was taken for testing.
- **Prioris:**

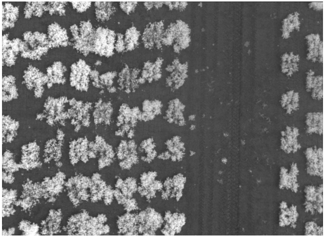
α shape $= 2$	μ mean (m) $= 0$
α rate $= 1$	μ sd (tau) $= 0.001$
β shape $= 2$	σ a $= 10$
β rate $= 1$	σ b $= 10$
δ shape $= 10$	δ rate $= 0.5$

- **Initialization:**
  - $\alpha = 2$
  - $\beta = 2$
  - $\delta_{1,2,3} = 30$

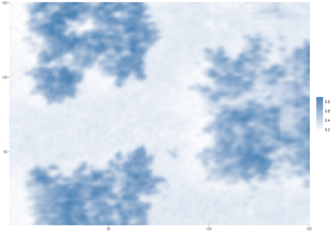
Image Segmentation



(a) Real colored image



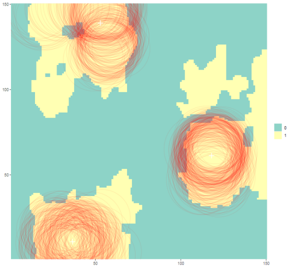
(b) Near-infrared image



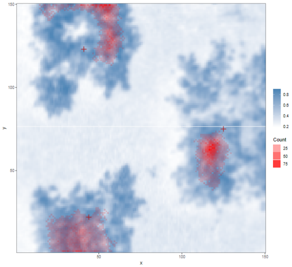
(c) Image frame

(d) Z initialization

Seed Locations



(a) Influence area estimated locations



(b) Seeds random walk heatmap

Image Segmentation Application Results

- **Good initialization:** Testing prioris may lead to a good first Z.
- **Fast Convergence:** After step 3001, parameters did not change drastically. The entire pipeline took $\sim$7 minutes for 10000 steps.
- **Accurate seeds:** Seed plots showed that seeds were well estimated as well as their influence area, but could get confused with background-within cases.

Conclusion

- **Key Conclusions:** The MWG pipeline makes inference in spatially complex HMRFs viable and robust.
- **Efficiency:** Estimation via Pseudoposterior balances computational cost and statistical precision.

Acknowledgements & Contact

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- **GitHub:** github.com/wescabral

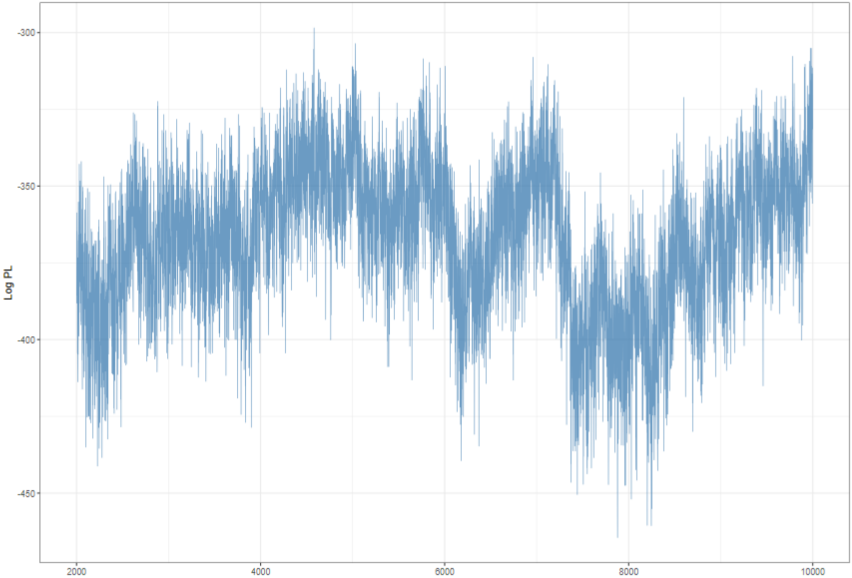
Questions & Discussion

Thank you for your attention! Feel free to reach out for questions.

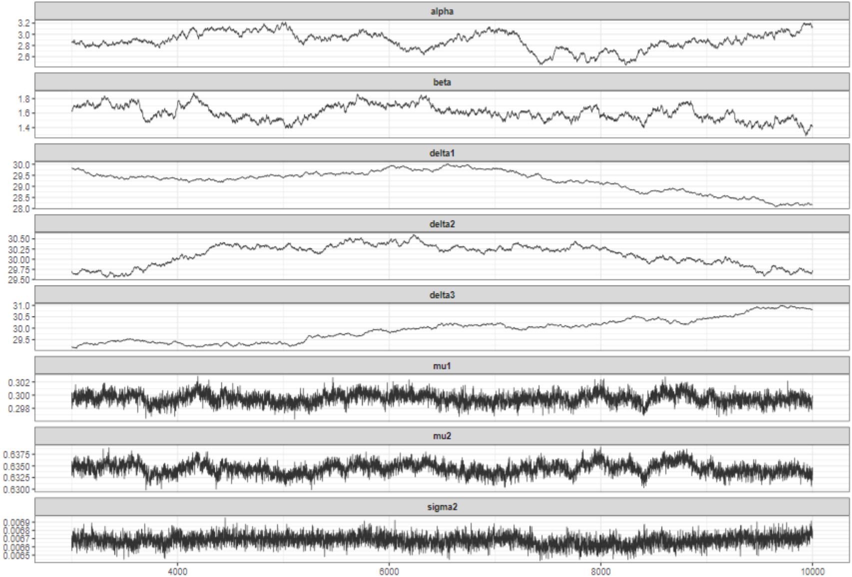
References

Aslahishahri, Masoomeh, Kevin G Stanley, Hema Duddu, Steve Shirtliffe, Sally Vail, Kirstin Bett, Curtis Pozniak, and Ian Stavness. 2021. "From RGB to NIR: Predicting of Near Infrared Reflectance from Visible Spectrum Aerial Images of Crops." In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 1312–22.

Parameters Convergence



Parameters Convergence



Parameters Convergence